

Project Report: Enhancing LLM Inference with GraphRAG

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Abstract

We present a GraphRAG system that enhances LLM inference by integrating knowledge graph retrieval with natural language question answering. The system operates on the Nobel Laureates dataset, translating user questions into Cypher queries executed against an embedded Kuzu graph database. We address two core challenges: improving Text2Cypher accuracy through dynamic few-shot exemplar selection, self-refinement with validation feedback, and rule-based post-processing; and optimizing system performance via LRU caching at the schema pruning and query generation stages. Our ablation study across eight enhancement configurations demonstrates that the combination of few-shot selection and post-processing achieves the best accuracy-latency trade-off, with 100% syntax validity, 90% semantic correctness, and average latency under 1.5 seconds. LRU caching reduces end-to-end latency from 1.77 seconds to 6.5 milliseconds on cache hits, a 270x improvement. The implementation and evaluation code are available at <https://github.com/dcgran/CS-E4780-project> [1].

Keywords

Large Language Models, Retrieval-Augmented Generation

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1 Introduction

Retrieval-Augmented Generation (RAG) systems enhance Large Language Model (LLM) inference by incorporating external knowledge sources to improve factual accuracy and reduce hallucinations. Traditional RAG approaches rely on semantic similarity search over unstructured text chunks, which works well for many tasks but struggles to capture the rich relational structure underlying complex domains such as scientific collaboration networks, enterprise knowledge bases, and multi-hop reasoning problems. GraphRAG addresses this limitation by integrating graph representations of knowledge into the retrieval process [2], enabling LLMs to reason over structured relationships and perform multi-hop queries that would be difficult to express through text-based retrieval alone.

This project implements a GraphRAG system using the Nobel Laureates dataset, which contains mentorship relationships spanning from the 16th century to 2021, enriched with biographical data from the official Nobel Prize API. The dataset is stored in Kuzu, a graph database that supports efficient pattern matching through the Cypher query language. The system pipeline consists of four

main stages: schema pruning to identify relevant graph structure, Text2Cypher generation to translate natural language questions into Cypher queries, query execution against the graph database, and response generation by the LLM based on retrieved results.

We address two primary objectives in this work. First, we enhance the Text2Cypher component to improve query generation accuracy through dynamic few-shot exemplar selection, self-refinement with validation feedback, and rule-based post-processing. Second, we optimize system performance through LRU caching and conduct comprehensive benchmarking to characterize latency across all pipeline stages.

Our evaluation includes an ablation study examining all combinations of Text2Cypher enhancements, quantifying accuracy-latency trade-offs. We measure both syntax and semantic correctness against a diverse test set. The results provide practical insights into which optimizations yield the best balance between quality and efficiency.

2 System Architecture

2.1 GraphRAG Pipeline

Our GraphRAG system implements a five-stage pipeline that transforms natural language questions into structured graph queries and generates contextually grounded responses. Figure 1 illustrates the complete workflow.

Stage 1: Schema Retrieval. The system begins by extracting the complete graph schema from Kuzu, including all node types, relationship types, and their associated properties. This schema provides a comprehensive structural description of the knowledge graph that will be used in subsequent stages.

Stage 2: Schema Pruning. Given a user question, the system uses an LLM to identify which subset of the schema is relevant to answering that question. This pruning step reduces the context size for downstream stages, improving both accuracy and latency.

Stage 3: Text2Cypher Generation. The pruned schema and user question are passed to the Text2Cypher component, which generates a syntactically and semantically valid Cypher query. This stage incorporates three enhancement mechanisms (detailed in Section 3.2): similarity-based few-shot exemplar selection, self-refinement with validation feedback, and rule-based post-processing.

Stage 4: Query Execution. The generated Cypher query is executed against the Kuzu database. Kuzu’s columnar storage and optimized query engine enable efficient pattern matching over the graph structure. The query results are returned as structured data (e.g., lists of nodes, relationships, or aggregated values).

Stage 5: Response Generation. Finally, an LLM synthesizes a natural language response based on the original question, the generated query, and the structured results from the database. This

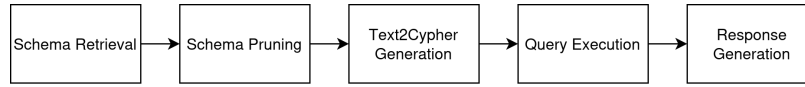


Figure 1: Flowchart depicting the complete workflow of the GraphRAG pipeline

stage translates the retrieved facts into a coherent, human-readable answer that directly addresses the user’s information need.

2.2 Knowledge Graph Schema

The Nobel Laureates knowledge graph represents prize winners, their institutional affiliations, and geographic metadata. The graph is constructed from the official Nobel Prize API, which provides biographical and institutional data for laureates from 1901 to 2021. Figure 2 illustrates the complete schema with all node types and properties.

The schema defines six node types: **Scholar** (laureate biographical data), **Prize** (award year, category, motivation, amount), **Institution** (affiliated organizations), and geographic entities **City**, **Country**, and **Continent**. Seven relationship types connect these entities: **WON** (Scholar→Prize), **BORN_IN** and **DIED_IN** (Scholar→City), **AFFILIATED_WITH** (Scholar→Institution), **IS_LOCATED_IN** (Institution→City), and the geographic hierarchy **IS_CITY_IN** (City→Country) and **IS_COUNTRY_IN** (Country→Continent).

This structure enables multi-hop queries such as: “How many Physics laureates were born in European countries?” or “Which institutions with Chemistry winners are in the United States?”

2.3 Component Interactions

The system follows a separation of concerns design with three main components: the core GraphRAG pipeline (question-to-answer flow), supporting modules (schema handling, few-shot retrieval, validation, caching), and benchmarking tools (accuracy and latency measurement). The core GraphRAG module orchestrates the pipeline stages, while the database manager handles connections and schema extraction, and DSPy components define structured prompts for pruning, query generation, and answer synthesis. This separation keeps the core logic simple and allows the evaluation scripts to run multiple configurations without modifying the pipeline. Strongly typed schema objects help guide the LLM toward valid outputs.

3 Implementation

3.1 Testbed Setup

Hardware Environment. Experiments were conducted on a machine with [TODO: e.g., “Intel i7-12700H” or “Apple M2”], [TODO: e.g., “16”] GB of memory, running [TODO: e.g., “Ubuntu 22.04” or “macOS 14”]. However, hardware specifications have limited impact on end-to-end latency in this system, as the LLM inference—which dominates execution time—is performed remotely via API calls rather than locally.

Software Environment. The system is implemented in Python 3.13 using the following key dependencies:

- **Kuzu** (v0.11.1): Embedded graph database supporting the Cypher query language, used for storing and querying the Nobel Laureates knowledge graph.
- **DSPy** (v3.0.1): Framework for building LLM-powered pipelines with structured prompts, used to orchestrate schema pruning, query generation, and answer synthesis.
- **Sentence Transformers** (v3.0.0): Provides the `all-MiniLM-L6-v2` embedding model for computing semantic similarity in KNN-based few-shot exemplar selection.
- **cachetools** (v6.1.0): Implements the LRU cache for storing query generation results.

LLM Configuration. All LLM inference is performed via the OpenRouter API using the `google/gemini-2.5-flash` model. This remote inference setup means that observed latencies are influenced by network conditions and API response times, which introduces variability across runs. We mitigate this by averaging results over multiple executions.

Test Query Dataset. We evaluate the system using a benchmark suite of 10 natural language questions designed to cover diverse query patterns:

- **Entity lookup:** “Who won the Nobel Prize in Physics in 2020?”
- **Multi-hop with constraints:** “Which scholars won prizes in Physics and were affiliated with University of Cambridge?”
- **Aggregation:** “How many Nobel Prizes have been awarded in Chemistry?”
- **Geographic filtering:** “List all Nobel Prize winners from Sweden.”
- **String matching:** “Find all scholars whose name contains ‘Einstein’.”
- **Adversarial queries:** Questions designed to stress-test the system, such as “List Nobel laureates born in Sweden and located in Paris” (requires correct join semantics) and “Which laureates winning literature prizes are affiliated with Stanford but not Cambridge?” (references non-existent prize category).

One query is duplicated to measure cache hit behavior. The adversarial queries help identify failure modes in query generation and schema understanding.

3.2 Task 1: Text2Cypher Enhancement

Dynamic Few-Shot Exemplar Selection. We implement dynamic exemplar selection using K-nearest neighbors (KNN) retrieval over a trainset of 15 curated question-query pairs covering diverse query patterns. At inference time, the user’s question is embedded using Sentence Transformers (`all-MiniLM-L6-v2`), and the $k = 3$ most similar examples are retrieved and included as few-shot context. This is implemented via DSPy’s `KNNFewShot` optimizer. We also support a static baseline that always selects the first k examples for comparison.

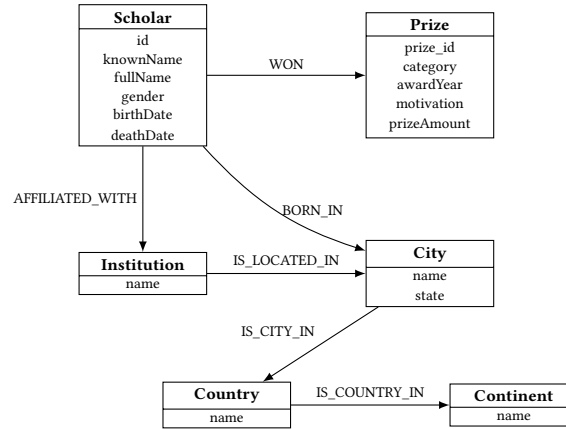


Figure 2: Knowledge graph schema showing node types and relationships in the Nobel Laureates dataset. The schema supports multi-hop queries across prize winners, their institutional affiliations, and hierarchical geographic locations.

Self-Refinement Loop. We implemented a self-refinement loop that improves the generated Cypher query until it passes a validation check. The validation works by running `EXPLAIN <query>` on the generated query, and the query is accepted if it does not raise an error. It also ensures that the answer is not a hard-coded string, e.g., `RETURN "Cannot answer"`. The self-refinement loop is run a maximum of 3 times, after which it rejects the query.

Rule-Based Post-Processing. After query generation, we apply regex-based transformations to convert string equality checks into case-insensitive `CONTAINS` patterns (e.g., `s.name = 'Einstein'` becomes `lower(s.name) CONTAINS 'einstein'`). This handles case mismatches and partial string matches on properties like name, fullName, and category.

3.3 Task 2: Caching and Performance

LRU Cache Design. We implement caching at two stages of the pipeline using the `cachetools.LRUCache` data structure with a maximum capacity of 128 entries each. The first cache stores schema pruning results, avoiding repeated LLM calls when the same question is asked against the same graph schema. The second cache stores generated Cypher queries, eliminating redundant query generation for previously seen question-schema combinations.

Cache keys are computed as a composite of question and schema hashes. Specifically, we compute the MD5 hash of the question text and the MD5 hash of the serialized schema, then concatenate the first 8 hexadecimal characters of each to form a key in the format `<q_hash>_<s_hash>`. This design ensures that identical questions produce cache hits only when the underlying schema is also unchanged, which is important for correctness when the graph structure evolves. The caches are scoped to each GraphRAG instance rather than shared globally, which prevents warm-cache effects from contaminating benchmark comparisons across different enhancement configurations.

Benchmarking Infrastructure. Timing data is collected using `time.perf_counter()` around each pipeline stage. [TODO: Specify number of runs per query (e.g., 10), warm-up runs excluded (e.g., 1), and aggregation method (e.g., mean/median).]

Memory Measurement. [TODO: Either describe memory profiling method (e.g., `tracemalloc`, `psutil` for Kuzu DB footprint) OR state "Memory profiling was not performed as LLM inference occurs remotely via API."]

4 Performance Evaluation

4.1 Task 1: Text2Cypher Accuracy and Latency

Using the benchmark suite described in Section 3.1, we evaluate 8 enhancement configurations covering entity lookup, multi-hop reasoning, boolean constraints, and schema disambiguation queries. Table 1 presents the ablation study showing accuracy-latency trade-offs. The Few-shot + Post-processing setting provided the best overall balance, achieving 100% success and syntax validity, 90% semantic correctness, and the lowest average latency (1.43 s). The All refinements configuration performed similarly but at a slightly slower average latency (1.61 s), which suggests that the additional refinement steps add overhead without improving the quality of the query’s semantics.

The baseline configuration achieved only 80% across all correctness metrics. Our enhancements improved semantic accuracy from 80% to 90%, a 12.5% relative improvement, while also reducing latency by over 3x compared to baseline. The self refinement only configuration was also inefficient, since while it maintained 100% success and validity with 90% semantic correctness, its average latency rose to 11.32 seconds. This is probably because of it running the expensive query generation multiple times.

All configurations except the baseline were able to produce a semantically correct query for 9 out of the 10 prompts. The only query that proved difficult was "List Nobel laureates born in Sweden and located in Paris." This is because the generated queries either connect Paris to the birth city or use a union instead of an intersection, and they don’t include the expected edge `is_located_in`.

4.2 Task 2: Performance and Efficiency

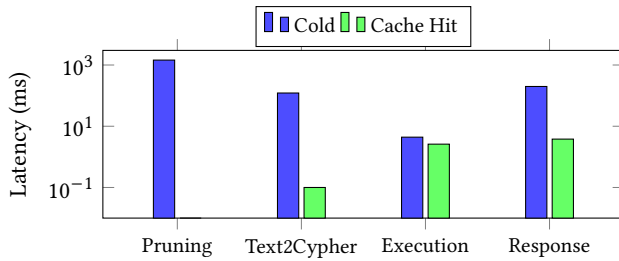
Table 2 shows the pipeline stage latency breakdown. It was generated using a separate script that aggregates the data from the

Table 1: Ablation Study - Accuracy vs Latency Trade-offs

Configuration	Syntax Correct (%)	Semantic Correct (%)	Avg Latency (ms)	Avg Iterations
Baseline (static few-shot)	80	80	4728	-
Only Few-shot selection	100	90	6494	-
Only Self-refinement	100	90	11316	1.0
Only Post-processing	90	90	1600	-
Few-shot + Post-processing	100	90	1430	-
Few-shot + Refinement	100	90	2043	0.9
Refinement + Post-processing	100	90	2189	1.0
All enhancements	100	90	1610	0.9

Table 2: Pipeline Stage Latency Breakdown (ms)

Stage	Cold	Cached
Schema Pruning	1447,3	0,0
Text2Cypher	121,7	0,1
Query Execution	4,4	2,6
Response Gen.	199,0	3,8
Total	1772,4	6,5

**Figure 3: Pipeline stage latency comparison: cold execution vs. cache hit (log scale). Caching eliminates schema pruning and query generation overhead.**

benchmarking suite. From the table, we can see that there is a significant difference between cold runs and cache hits. Schema pruning, which is normally the slowest step, drops from about 1.45 seconds to essentially zero when cached. Query generation shows a similar pattern, falling from 121.7 ms to 0.1 ms. Response generation also becomes much faster, though not fully eliminated, and query execution sees only a small improvement. Together, these changes reduce end-to-end latency from 1.77 seconds to just 6.5 ms on cache hits.

In cold runs, schema pruning is the clear bottleneck, responsible for over 80% of total latency. On cache hits, pruning and query generation latency effectively disappear. Figure 3 visualizes this breakdown, showing the dramatic reduction from cold to cached execution. Cache hit rates averaged 10% in our benchmark due to the diverse query set; production workloads with repeated queries would see substantially higher rates.

5 Discussion

The key insight from our ablation study is that lightweight techniques—few-shot selection and post-processing—outperform expensive self-refinement while achieving comparable accuracy. Self-refinement’s validation loop is most valuable for complex multi-hop queries, but its latency cost (11+ seconds) makes it impractical as a default. Post-processing provides robust case-insensitive matching at negligible cost, which is critical since entity names often differ between user questions and database values.

A consistent challenge was ambiguous geographic queries. The test case “List Nobel laureates born in Sweden and located in Paris” failed across configurations because “located” could mean birth location, institution location, or residence—highlighting that Text2Cypher accuracy is bounded by question clarity.

For production deployments, we recommend Few-shot + Post-processing with LRU caching. This achieves 90% semantic accuracy at minimal latency, with cache hits providing 270x speedup for repeated queries.

6 Conclusion

We presented a GraphRAG system addressing Text2Cypher accuracy and performance. For Task 1, we implemented KNN-based few-shot selection, EXPLAIN-based self-refinement, and rule-based post-processing. Our ablation study showed that few-shot selection combined with post-processing achieves the best accuracy-latency trade-off. For Task 2, LRU caching at schema pruning and query generation stages reduces latency from 1.77 seconds to 6.5 milliseconds on cache hits. The practical takeaway is that lightweight post-processing can substitute for expensive iterative refinement, and caching is essential for interactive GraphRAG systems.

References

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- [2] Haoyu Han, Yu Wang, Harry Shomer, Kai Guo, Jiayuan Ding, Yongjia Lei, Mahantesh Halappanavar, Ryan A. Rossi, Subhabrata Mukherjee, Xianfeng Tang, Qi He, Zhigang Hua, Bo Long, Tong Zhao, Neil Shah, Amin Javari, Yinglong Xia, and Jiliang Tang. 2025. Retrieval-Augmented Generation with Graphs (GraphRAG). arXiv:2501.00309 [cs.LG] <https://arxiv.org/abs/2501.00309>